Phase 2: Normalize + Model

Mapping messy, inconsistent XBRL tags onto one clean, typed representation of a company-year.

What this phase does

Two companies reporting the same economic fact -- cost of goods sold, say -- often use different XBRL tags for it. normalize.py holds a table of canonical line item -> candidate tags, tried in order, and assembles the result into typed pydantic models (IncomeStatement, BalanceSheet, CashFlowStatement) where every field is Decimal or None -- never float, and never a fabricated zero standing in for "we don't know."

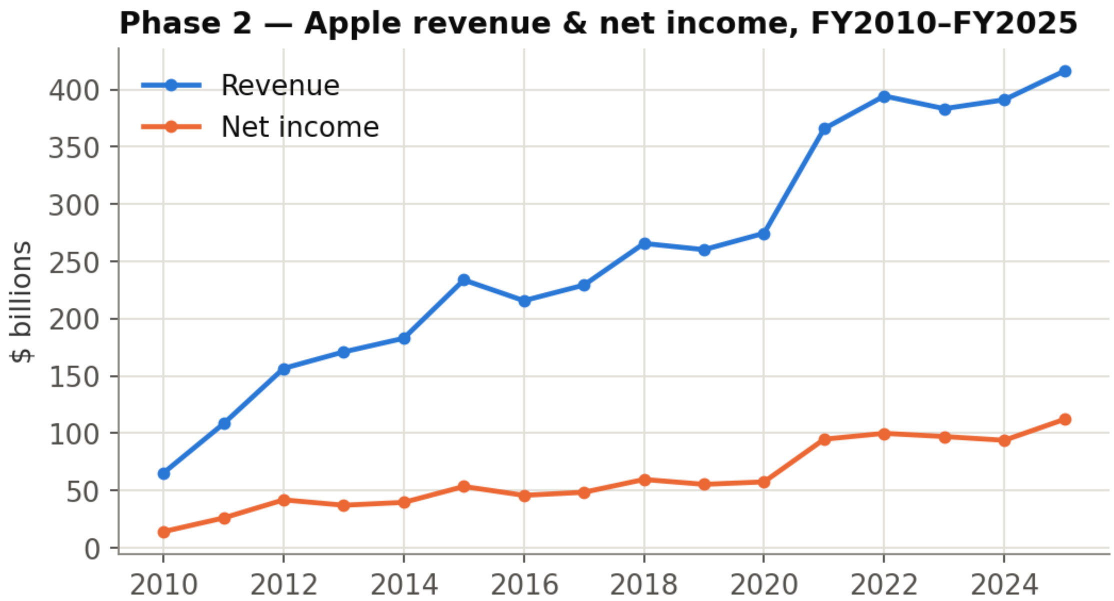
A real bug this phase caught: SEC's fy/fp field is not what it looks like

It's tempting to assume a fact's "fy" field tells you which fiscal year that number covers. It doesn't -- fy/fp describe the filing's own fiscal focus, not the period an individual fact covers. A single 10-K reports 2-3 years of comparative figures, and every one of those rows is stamped with that filing's own fy. Filtering on fy == target_year silently pulled the wrong year's numbers in an early version of this tool; the bug only surfaced when printing a "FY2023" model against real Apple data showed FY2021's date range. The fix: anchor the target period by finding the row with the latest end-date within that fy/fp group, then match every other concept purely by (start, end) dates -- which also correctly picks up a restated value filed later, since a restatement shows up as a comparative row in a subsequent 10-K carrying that later filing's own fy.

Real cross-company tag divergence found while hardening this phase

- Walmart and Coca-Cola never tag a total "Liabilities" concept at all, in any filing -- their balance sheets simply don't present that subtotal as its own XBRL fact.
- Johnson & Johnson hasn't tagged "OperatingIncomeLoss" directly since 2014.
- Microsoft tags Selling & Marketing and General & Administrative expense as two separate concepts, with no combined SG&A tag at all.
- JNJ and Coca-Cola tag common stock par value and additional paid-in capital as separate concepts; JNJ doesn't tag APIC at all in recent filings.

None of these are bugs in the filing -- they're legitimate differences in how each company's 10-K is structured. normalize.py handles them with derivation: sum the parts when a combined tag is missing (SG&A = Selling + G&A), or take the difference of two totals that are always tagged (Liabilities = Assets - Equity). A component that's genuinely absent (not every company tags additional paid-in capital) is treated as 0 only when deriving a sum -- never silently assumed elsewhere.



Apple's revenue and net income, normalized from raw XBRL facts across 16 fiscal years.

KEY TAKEAWAY

The typed model's real value isn't the numbers it successfully fills in -- it's the fields it leaves as None. A missing value that propagates cleanly through every downstream calculation is honest; a missing value silently treated as zero would quietly corrupt every ratio and check built on top of it.